

Activity 11

Writing a c program to finding the sum of the squares of digits of number user has input.

```
#include <stdio.h>
int main(){
    int num, choice, temp, digit, sum;
    printf("Enter the number you wanted to be calculated");
    scanf("%d", &num);

    do{
        printf("\n-----Menu-----\n");
        printf("1.find the sum of squares of the digits.\n");
        printf("2.Exit\n");
        scanf("%d", &choice);

        switch(choice){
            case 1:
                temp = num;
                sum = 0;
                while(temp > 0){
                    digit = temp % 10;
                    sum += digit*digit;
                    temp /= 10;
                }
                printf("sum of squares of digit of a number:%d", sum);
                break;

            case 2:
                printf("Exiting program.\n");
                break;

            default:
                printf("Invalid choice");
                break;
        }
    }while(choice != 2);
    return 0;
}
```

Compilation and execution

```
(kali㉿kali)-[~]
└─$ vi newd4.c

(kali㉿kali)-[~]
└─$ gcc newd4.c -o newd4
```

Input	Expected OUTPUT	Actual OUTPUT	Result
12	5	5	PASS
13	10	10	PASS
17	50	50	PASS
19	82	82	PASS

Algorithm

- 1.START
2. define num, digit, temp, sum as int.
3. start do while loop.
4. insert switch case inside them.
5. make 2 choice case one for printing result , and another for exiting.
6. STOP